

Transformations, Groups and Geometry

UNSW MATH3511 25T3

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0 Revision

Here is a list of concepts you should know from high school.

- Vertically opposite, alternating, corresponding and co-interior angles.
- Interior angles of a triangle sum to π ; exterior angle equals sum of opposite interior angles.
- Congruent and similar triangles.
- Isosceles and equilateral triangles.
- Sine rule: $\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$.
- Cosine rule: $a^2 = b^2 + c^2 - 2bc \cos \alpha$. Pythagoras' theorem if $\alpha = \frac{\pi}{2}$.
- Three non collinear points defines a unique circle.
- A quadrilateral is cyclic if its vertices lie on a circle.

1 Triangle Geometry

1.1 Signed Lengths and Orientation

A line through two points A and B is denoted AB , typically *in that direction*. For a point P on the line AB , if P lies between A and B then the ratio $\frac{AP}{PB} > 0$. But if P is outside A and B then $\frac{AP}{PB} < 0$. If P and A are the same point, then the ratio is zero; if P and B are the same, then the ratio is ∞ .

An **altitude** of a triangle is a line segment through a given vertex and perpendicular to the line containing the side opposite the vertex.

In a triangle, the orientation is called **positive** if the vertices are taken anti-clockwise in the order given. For an oriented triangle, we take directed lengths to match the orientation: AB, BC and CA are positive in $\triangle ABC$. We extend this convention to area and angle:

- The area of a triangle is positive if the triangle is positively oriented; otherwise it is negative.
- Positive angles go anticlockwise. If edges are signed, we need to account for that. We take $\angle ABC$ as positive if the sense runs anticlockwise from BA to BC when both are the same sign; if they are opposite signs, we change the sign of the angle to compensate.

We denote the sign of the area as $\triangle ABC = \frac{1}{2}|AB| \cdot |BC| \cdot \sin \angle ABC$.

1.2 Intersections

A **centroid** is the intersection of the medians while a **orthocetnre** is the intersection of altitudes. It can be shown that the medians are concurrent. The medians and altitudes are examples of **cevians**.

Definition. (Cevian) In any triangle a line segment joining a vertex to a non-vertex point on the opposite side, possibly extended, is called a cevian.

Theorem. (Ceva's Theorem (1678)) In $\triangle ABC$ the cevians AD , BE and CF (where D is on BC , E is on AC and F is on AB) are concurrent if and only if

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1.$$

Corollary. In $\triangle ABC$ the cevians AD , BE and CF are concurrent if and only if

$$\frac{\sin ACF}{\sin FCB} \cdot \frac{\sin BAD}{\sin DAC} \cdot \frac{\sin CBE}{\sin EBA} = 1,$$

taking proper account of orientations of the angles.

1.3 Collinearity

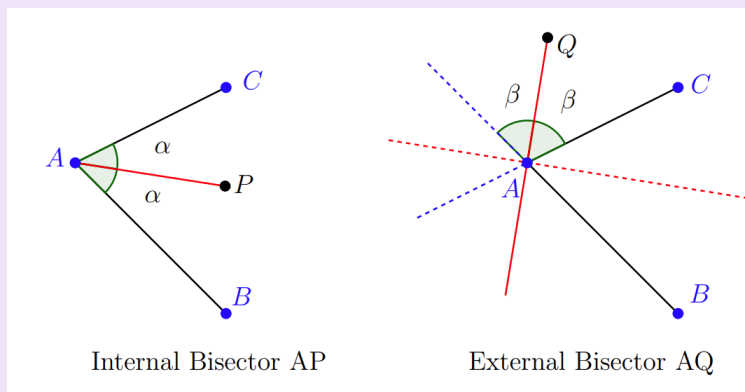
Theorem. (Menelaus' Theorem) In $\triangle ABC$ let D , E and F be on side BC , CA , AB (or their extensions) respectively. Then D , E and F are collinear if and only if

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = -1.$$

Theorem. (Pappus' Theorem) Suppose points A, C, E lie on one line and B, D, F on a second line, where AB, CD and EF form a triangle. Let $L = AB \cap DE$, $M = CD \cap FA$ and $N = EF \cap BC$. Then L, M and N are collinear.

1.4 Angle Bisectors

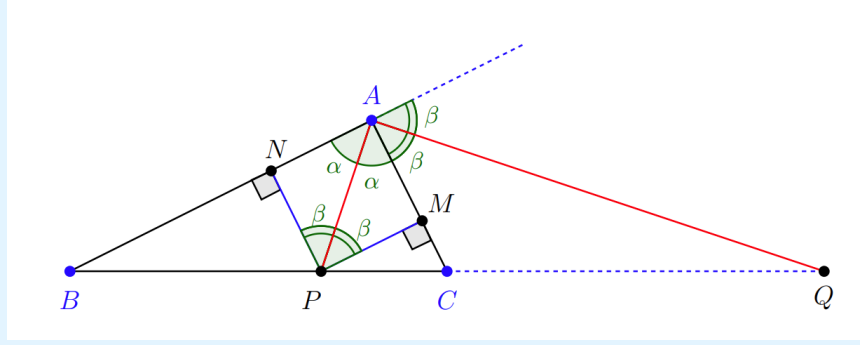
Definition. (Bisectors) In the image below, the line AP in the left hand diagram is the **internal bisector** of $\angle BAC$: $\angle BAP = \frac{1}{2}\angle BAC$ and the line AQ in the right hand diagram is the **external bisector** of $\angle BAC$: $\angle CAQ = \frac{1}{2}(\pi - \angle BAC)$, that is it bisects angle between AC and the extension of the line AB .



Theorem. (Angle Bisector Theorem) Consider $\triangle ABC$, and its angle bisector as below.

a) If the interior bisector of $\angle BAC$ meets BC at P then $\frac{BP}{PC} = \frac{AB}{CA}$.

b) If the exterior bisector of $\angle BAC$ meets BC at Q then $\frac{BQ}{QC} = -\frac{AB}{CA}$.



Theorem. (Incentre Theorem) The internal bisectors of a triangle are concurrent.

The point of concurrence is called the **incentre**. This point is equidistance from the three sides, so a circle centred here and tangent to the edges is the largest circle that can fit inside the triangle. This circle is called the **incircle**.

Theorem. (Excentre Theorem) The external bisectors of any two angles of a triangle are concurrent with the internal bisector of the third angle.

The points of concurrence (there are three of them) are called the **excentres**. Again, each excentre is the same distance from the three sides, and so circles centres at the excentres with that distance as radius (the **excircles**) are tangent to the sides of the triangle, one internally, two externally.

Theorem. In $\triangle ABC$, let the internal bisectors at angles A, B and C meet the opposite sides at P, Q and R respectively and the external bisectors meet the opposite sides at P', Q' and R' respectively. Then $\{P, Q, R'\}$, $\{P, Q', R\}$ and $\{P', Q, R\}$ are triples of collinear points.

1.5 Area Formulas

Let r be the radius of the incircle. Then

$$\triangle ABC = \frac{1}{2}r(a + b + c) = sr,$$

where $s = \frac{1}{2}(a + b + c)$ is the semiperimeter.

We can get similar results from the excircles. Let r_a be the radius of the excircle centred at J_a . Then

$$\triangle BCJ_a = \frac{1}{2}r_a \cdot a, \quad \triangle ABJ_a = \frac{1}{2}r_a \cdot c, \quad \triangle ACJ_a = \frac{1}{2}r_a \cdot b,$$

since r_a is the distance to each edge. Thus,

$$\triangle ABC = \frac{1}{2}r_a(b - a + c) = (s - a)r_a.$$

Then similarly, we have

$$\triangle ABC = (s - b)r_b = (s - c)r_c.$$

After some trigonometry and rearranging to solve for r , we get

$$r = \sqrt{\frac{(s-a)(s-b)(s-c)}{s}}.$$

Then from the previous expression we get **Heron's formula**:

$$\Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}.$$

Then we get the following results,

$$r_a = \sqrt{\frac{s(s-b)(s-c)}{s-a}} \quad \text{and} \quad \frac{1}{r} = \frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c}.$$

1.6 Circles Associated to Triangles

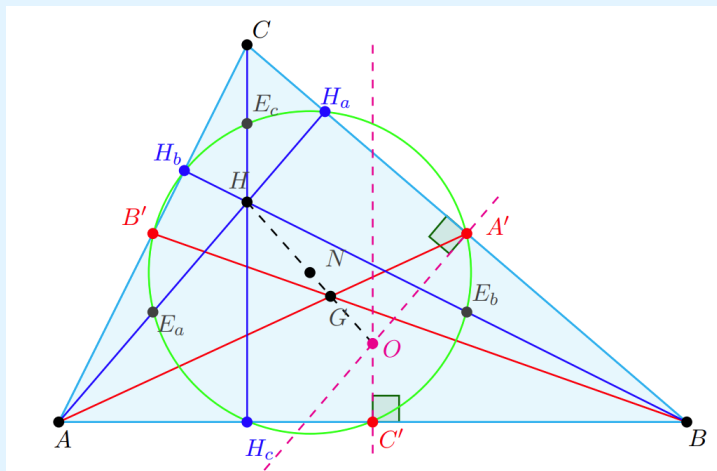
The **circumcircle** goes through the three vertices of a triangle and has centre the **circumcentre**, the intersection of the three perpendicular bisectors of the sides.

Suppose for a circle $\triangle ABC$ with circumcentre O . Let $OA = OB = OC = R$ be the **circumradius**. Then $\angle AOB = 2\angle C$ and so $\angle C = \angle FOB$. Thus $R \sin C = FB = \frac{c}{2}$. Then similar results for the other sides gives the **extended sine rule**:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

Lemma. The circumcentre of a triangle is the orthocentre of its median triangle.

Theorem. (Nine Point Circle) For $\triangle ABC$, let A', B' and C' be the midpoints of the three sides and H_a, H_b and H_c be the feet of the altitudes. Let H be the orthocentre and E_a, E_b and E_c be the midpoints of the lines from H to the three vertices. Also let O be the circumcentre and R the circumradius. Then if N is the midpoint of the line from H to O , the nine points $A', B', C', H_a, H_b, H_c, E_a, E_b$ and E_c lie on the circle centre N of radius $\frac{R}{2}$.



Theorem. (Feuerbach's Theorem) The nine-point circle is tangent to the incircle and all three excircles.

The point where the incircle and the nine-point circle meet is called the **Feuerbach point**. It is collinear with the incentre and the nine-point centre.

Theorem. (The Euler Line) In any triangle, the orthocentre H , nine-point centre N , circumcentre O and centroid G are collinear. Furthermore N is the midpoint OH and $OG : GH = 1 : 2$.

The line these points lie on is called the **Euler Line**.

Theorem. (Simson Line) Let P be any point on the circumcircle of $\triangle ABC$. Then the feet of the perpendiculars from P to each side of $\triangle ABC$ are collinear.

Proposition. The three Euler points uniquely define a triangle.

1.7 The Cross Ratio

Theorem. Let $\triangle ABC$ have points D, E, F on the sides opposite A, B, C respectively and suppose the cevians AD, BE and CF are concurrent at a point P inside $\triangle ABC$. Let Q be a point on BC . Then F, E and Q are collinear if and only if $\frac{BD \cdot CQ}{CD \cdot BQ} = -1$.

Definition. (Cross Ratio) Let A, B, C, D be any four points. The cross ratio $(A, B; C, D)$ is defined by

$$(A, B; C, D) = \frac{AC}{BC} \div \frac{AD}{BD} = \frac{AC \cdot BD}{BC \cdot AD}.$$

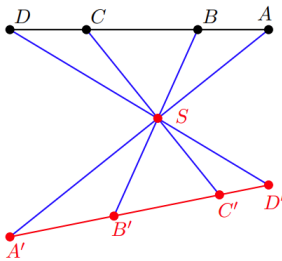
Note that changing the order of the points in $(A, B; C, D)$ may change the value of the cross ratio. However

$$(A, B; C, D) = (B, A; D, C) = (C, D; A, B) = (D, C; B, A),$$

so there are at most 6 different values of the cross ratio.

Definition. (Harmonic) A set of collinear points $\{A, B, C, D\}$ for which $(A, B; C, D) = -1$ are said to be harmonic, and C, D are called the harmonic conjugates of A, B .

Theorem. In the diagram below, we have



$$(A, B; C, D) = (A', B'; C', D') = \left(\frac{\sin \angle ASC}{\sin \angle BSC} \right) \cdot \left(\frac{\sin \angle BSD}{\sin \angle ASD} \right).$$

The sets $\{A, B, C, D\}$ and $\{A', B', C', D'\}$ are said to be in perspective from S , or one set if projected from S . Our result can then be stated as the cross ratio of collinear points is a projective invariant.

Theorem. Suppose that in $\triangle ABC$, X and Y are points on AB such that CX is the internal bisector of $\angle ACB$ and CY its external bisector. Then $(A, B; X, Y) = -1$, i.e. the four points are harmonic.

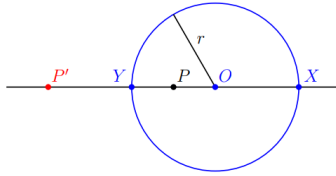
Conversely, given points X and Y on side AB of $\triangle ABC$ with CX orthogonal to CY , if $(A, B; X, Y) = -1$ then CX is the internal and CY the external bisector of $\angle ACB$.

2 Circle Geometry

A circle with centre at a point O and with radius r will be denoted $\mathcal{C}(O, r)$.

2.1 Inversion

Theorem. For circle $\mathcal{C}(O, r)$, let $P \neq O$ be any point and X, Y the intersection of $\mathcal{C}$ and the line through O and P as below. Then $(X, Y; P, P') = -1$ if and only if $OP \cdot OP' = r^2$.



Definition. (Inverse) Let $\mathcal{C}(O, r)$ be a circle. Given a point P distinct from O we can define a point P' called its inverse in $\mathcal{C}$ as follows: on the line OP the point P' is such that $OP \cdot OP' = r^2$.

Definition. The map $T : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^2 \setminus \{0\}$ sending each P to its inverse in $\mathcal{C}$ is called inversion. We call O the centre of inversion and r the radius of inversion.

Simple properties of an inversion, T , are immediate:

- T pointwise fixes the circle (if P is on the circle $T(P) = P$);
- T maps the inside of the circle to the outside and vice versa;
- T is an involution, that is $T^2(P) = P$.

Theorem. If a circle $\mathcal{C}(O, r)$ inverts A, B to A', B' respectively, then

$$A'B' = \frac{r^2 AB}{OA \cdot OB}.$$

Theorem. Let A, B, C, D be 4 points all distinct from O . If circle $\mathcal{C}(O, r)$ inverts A, B, C, D to A', B', C', D' respectively, then

$$(A, B; C, D) = (A', B'; C', D').$$

2.2 Poles and Polar

Definition. (Polar and Poles) Let $\mathcal{C}(O, r)$ be a circle. For a point $P \neq O$ its polar, p , is the line through its inverse point P' orthogonal to OP' .

Conversely, given a line ℓ not through O its pole, L , is the inverse point to the point L' closest to O .

Theorem. If P is external to $\mathcal{C}(O, r)$, then its polar, p , is the line through the points where the tangents from P meet the circle.

Corollary. There is a ruler and compass construction of the tangents to a circle from an arbitrary external point.

Theorem. For $\mathcal{C}(O, r)$ and a point P that is neither O nor a point on the circle, suppose a line ℓ through P intersects $\mathcal{C}$ at distinct points X and Y .

Then if E is the intersection of ℓ and the polar of P , ℓ is divided harmonically by X, Y, E and P : i.e. $(X, Y; E, P) = -1$.

Conversely, if E on ℓ is such that $(X, Y; E, P) = -1$, then the line perpendicular to OP through E is the polar of P .

Theorem. Let $\mathcal{C}(O, r)$ be a circle. If ℓ is any line not passing through O , its inverse is a circle through O (minus the point O).

Conversely, if $\mathcal{C}'$ is a circle through O , its inverse is a line perpendicular to the diameter of $\mathcal{C}'$ through O .

Corollary. A circle $\mathcal{C}'$ passing through O is inverted in circle $\mathcal{C}(O, r)$ to a line parallel to the tangent to $\mathcal{C}'$ at O .

Theorem. For $\mathcal{C}(O, r)$ and $L \neq O$, let ℓ be its polar. Then for any point Q on ℓ , the polar of Q , q , passes through L .

Conversely, for ℓ a line not passing through O with pole L , any line q through L has pole Q on ℓ .

Corollary. For given circle $\mathcal{C}(O, r)$, let (P, p) and (Q, q) be two pole/polar pairs, where $N = p \cap q$ exists. Then the line PQ is the polar, n , of N and the pole of PQ is N .

2.3 Intersection of Circles

In general, two curves intersecting at a point P are said to be **orthogonal** if their tangents at P are orthogonal.

Theorem. Two intersecting circles are orthogonal if and only if the angles in opposite segments are complementary.

Theorem. Let A, B, C, D be a harmonic set, $(A, B; C, D) = -1$. The two circles with diameters AB and CD are orthogonal.

Conversely, if two circles with centres O_1 and O_2 are orthogonal, and the line O_1O_2 intersects one circle at A and B and the other at C and D then $(A, B; C, D) = -1$.

2.4 Inverting Circles and Lines

Given a circle $\mathcal{C}(O, r)$, we saw in a previous result that a circle through O is inverted to a line not through O , and conversely. It is clear that a line through O is inverted to a line through O , namely the same line.

Theorem. Let $\mathcal{C}(O, r)$ be a circle and $\Sigma(\Omega, \rho)$ a second circle not passing through O . Under inversion in $\mathcal{C}$, Σ is inverted to another circle not through O .

Theorem. Let $\mathcal{C}(O, r)$ be a circle and $\Sigma(\Omega, \rho)$ a second circle meeting $\mathcal{C}$ orthogonally. Then Σ is inverted to itself in $\mathcal{C}$.

When inverting in a circle $\mathcal{C}(O, r)$, we avoided inverting O itself. However, we can imagine adding a **ideal point**, ∞ “at infinity” to the plane, to which O is mapped by inversion in a circle centred at O .

This extended plane has many nice properties, which will explore later, but here we note that we can think of a line in the plane as a circle through ∞ in the extended plane. Then we can sum up our results on inversion as:

Theorem. In the extended plane, inversion in a fixed circle maps a circle to a circle.

Clearly any translation or rotation about a fixed point will send a line to a line and a circle to a circle. It follows that the most general transformation that maps a circle or line to a circle or line is in fact a composition for a translation, a rotation and an inversion.

Thinking of $\mathbb{R}^2$ as $\mathbb{C}$, inversion in the circle $|z-c| = r$ is the map $z \mapsto c + \frac{r^2}{|z-c|^2}(z-c)$ and is the composition of conjugation and a special case of a **linear fractional transformation**,

$$z \mapsto \frac{az + b}{cz + d}.$$